RESEARCH BRIEF

Brain tumor detection system and paper: what is new, and what comes next

Single-slice 2D MRI tumor segmentation, type classification and measurement, compared with research published 2024–2026

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In short

The system takes **one 2D MRI slice** (JPG or PNG) and returns the tumor **outline, type, size in millimetres and side of the brain**, with a warning when the outline is uncertain. The paper evaluates this **on patients the model never saw**, after removing copied images between the two public datasets.

What is new: one model that segments, classifies *and measures*, checked against expert masks on patient-level splits; measurement agreement (Bland–Altman) and side agreement reported; failure cases, calibration and an uncertainty flag reported; runs on a normal CPU in about 0.15 s.

What is not new: the network (ResNet34 U-Net) and the observation that the Kaggle and figshare benchmarks leak. In September 2026 another group published a large leakage audit for *classification*, so the paper must cite it and present its own leakage control as applied to *segmentation and measurement*.

Most important next steps: an external test set, a fair segmentation baseline (U-Net, nnU-Net), several training seeds, and real pixel spacing from DICOM files.

1 Where the field stands (2024–2026)

1.1 Classification on public 2D datasets is at the ceiling, and the benchmarks leak

- Most studies on the figshare (Cheng) and Kaggle (Nickparvar) datasets report 97–100% accuracy for four-class classification.
- **Vangala et al. (arXiv, September 2026)** [1] audited these datasets. In the current Kaggle test split (version with 1600 test images; our study used the earlier 1311-image version), 28.8% of test images (460 of 1600) have a near-copy in training. Of the test images that can be traced to a figshare patient, 95.5% share a patient with training. File headers alone (size, JPEG tables, no anatomy) reach 0.959 balanced accuracy for tumor vs no tumor. Removing leaked images changed accuracy by only −0.009, so high scores mainly show that the task on these data is easy, not that the model is clinically ready.
- A second audit of Kaggle version 2 [2] reports a copy in training for 79% of test images. A Research Square preprint [3] reports patient-level leakage for all traceable test images.
- All of these audits are about **classification only**. None of them evaluates segmentation, measurement or side.

1.2 2D segmentation papers report high Dice, often on image-level splits

- Recent figshare segmentation papers report Dice of about 0.90–0.93 (for example an attention U-Net with EfficientNetB4 encoder, Scientific Reports 2025 [4]). Many use random 80:20 or 90:10 image splits, so slices of the same patient appear in training and test, which inflates Dice.
- Joint segmentation + classification models exist (for example a densely connected network, 2026 [5]), but they rarely report tumor size, side, calibration or failure analysis.

1.3 Foundation models (SAM, MedSAM, SAM 2)

- A 2026 benchmark across five cancer types [6] found that fine-tuned Medical SAM 2 is strongest overall, but on brain tumors (BraTS) the supervised Swin UNETR ranked first. Zero-shot SAM models are clearly weaker on brain MRI.
- In our target journal (JIIM, 2026) [7], a five-centre Japanese study fine-tuned SAM and MedSAM with low-rank adaptation (LoRA) on **2D glioma slices** and reached Dice of about 0.91–0.92 on an external test set. This is the closest recent work to ours and should be cited and compared.

1.4 Automated tumor measurement

- AutoRANO (2019) [8] and later glioblastoma work (2023) [9] compute bidimensional RANO diameters and volumes automatically, but from **3D multi-sequence MRI** and for gliomas only.
- A 2026 review [10] reports that no machine-learning tool had yet been validated against the updated RANO 2.0 criteria.
- Measurement from a **single 2D image without pixel spacing**, as uploaded by a patient or general doctor, has received very little attention.

1.5 3D challenges and generalization

- BraTS 2025 [11] now covers adult glioma, meningioma, meningioma radiotherapy, metastases and paediatric tumors on 3D T1, T1-contrast, T2 and FLAIR. Leading teams report that the choice of backbone has plateaued and that workflow design (ensembles, subtype-aware training) matters more [12].
- Models lose reliability on data from other hospitals and scanners because of differences in hardware, protocols and intensities [13]. External validation is now expected by reviewers.

2 What our system and paper add

The table compares typical recent work with this study. Rows marked in blue are where this study goes beyond the typical paper on the same public data.

Aspect	Typical 2D papers on figshare / Kaggle	Leakage audits 2026 [1–3]	This study
Output	Class label, sometimes a mask	Class label	Outline + type + longest and perpendicular diameter + area + side + uncertainty flag
Data split	Random image-level	Deduplicated test split	Patient-level (figshare) + removal of Kaggle copies of held-out slices + separate copy-free subset
Segmentation evaluation	Dice, often image-level	Not done	Dice 0.750 (95% CI 0.657–0.838), median 0.863, HD95, per tumor type, patient-level bootstrap CIs
Measurement accuracy	Rarely reported	Not done	Longest diameter vs expert masks: median error 3.6 mm; Bland–Altman analysis
Side of the brain	Not reported	Not done	Radiological convention; sagittal slices detected and excluded; 100% agreement when the outline overlaps the tumor
Reliability	Accuracy only	Balanced accuracy, calibration recommended	Calibration (ECE 0.023), low-confidence flag, worst-case analysis
Reporting standard	Rarely	—	CLAIM checklist completed, TRIPOD+AI cited
Open material	Sometimes code	File lists, splits	Code, patient split, result for every test image
Practical use	Notebook	—	Web application, CPU only, 0.15 s per image

CI, confidence interval; HD95, 95th-percentile Hausdorff distance; ECE, expected calibration error.

2.1 Our main results at a glance

Measure	Result
Segmentation, 295 slices from 24 unseen patients	Mean Dice 0.750; meningioma 0.942, pituitary 0.801, glioma 0.634; 85.8% of slices with Dice ≥ 0.5
Longest diameter	Median absolute error 3.6 mm; 95% limits of agreement -30.1 to 29.9 mm (wide, because of a few large glioma errors)
Side	90.4% agreement overall; 100% when the outline overlaps the tumor
Four-class classification (Kaggle test)	98.9% (all 1311 images); 98.8% on the 690 images without a copy in training
Tumor detection	99.77%, versus 93.7% for the earlier 3D detection network

> **How to state the contribution (suggested wording for the paper)**
>
> "Leakage in the public brain tumor MRI benchmarks has recently been documented for classification [Vangala et al. 2026]. We extend leakage-controlled evaluation to segmentation and measurement: on patient-level splits, a single model outlines, classifies and measures the tumor on one 2D slice, and we quantify diameter and side agreement against expert masks, which classification-only studies cannot assess."
>
> **Action:** add references [1] and [7] to the JIIM manuscript before submission. Reviewers who know these papers will otherwise question the novelty claim.

3 Weak points reviewers will raise

Weak point	Why it matters	How to fix it
No external test set	Both datasets are public, overlapping and from few centres	Test on slices from another source with expert masks, e.g. 2D slices extracted from BraTS (glioma, meningioma) or a local hospital set
No segmentation baseline	Dice cannot be compared fairly with other methods	Finish the classic U-Net run (stopped by low memory) and add a 2D nnU-Net on the same split [14]
Small test set	24 test patients, only 3 with pituitary tumor	Patient-level cross-validation (5 folds) over all 233 patients
Single seed, no tuning	Result may depend on chance	3 seeds; report mean ± standard deviation
Size without pixel spacing	Scale estimated from head width, about 15% error	Accept DICOM input and read Pixel Spacing; validate on data with known spacing
Risk score not validated	Could mislead users	Keep it out of the paper (already done) or validate it with clinicians
Glioma failures	Skull-base and posterior-fossa gliomas are missed	More glioma data, 2.5D input (neighbouring slices), FLAIR when available

4 What we want to bring next (roadmap)

Phase 1 — before or during review (1–2 months)

1. Cite the 2026 leakage audits and the JIIM LoRA-SAM paper; sharpen the novelty statement as above.
2. Run the classic U-Net baseline and a 2D nnU-Net on the identical split; add 3 seeds and 5-fold patient-level cross-validation.
3. Add a header-only baseline (as recommended in [1]) to show how much of the Kaggle accuracy comes from file properties rather than anatomy.

4. Release the code with a Zenodo DOI and licence; add the deduplicated split files.

Phase 2 — second paper (3–6 months)

1. **External validation:** extract axial and coronal slices with masks and known pixel spacing from BraTS 2023–2025 (glioma, meningioma) and test segmentation, measurement and side without retraining.
2. **Foundation model comparison:** fine-tune MedSAM / SAM 2 with LoRA on the same patient-level split and compare accuracy, speed and CPU memory with our model.
3. **DICOM support with real pixel spacing**, so the reported size in millimetres is exact rather than estimated.
4. **2.5D input** (the slice plus its neighbours) to reduce glioma failures when a volume is available.
5. **Robustness to real-world uploads:** test on photographs of film and phone screenshots, which no recent study has evaluated, and add a quality check that rejects unusable images.

Phase 3 — clinical direction (6–18 months)

1. **Reader study with radiologists in Bangladesh** (with ethics approval): does the outline and measurement save time or change agreement between readers?
2. **Fix the 3D pipeline:** train the segmentation head of the 3D model on BraTS so that NIfTI volumes also get outlines and volumes.
3. **RANO / RANO 2.0-style bidimensional measurement** on the slice with the largest tumor extent, compared with radiologist measurements.
4. **Uncertainty-aware triage:** use calibrated probabilities and the low-confidence flag to send uncertain cases to a specialist.
5. **Report in Bengali and English** for patients and general doctors, clearly marked as decision support, not diagnosis.

5 Recommended order of work

Priority	Task	Effort	Effect on acceptance
1	Cite 2026 leakage audits and JIIM LoRA-SAM paper; adjust novelty text	1 day	High: avoids a novelty objection
2	U-Net and 2D nnU-Net baselines on the same split	1–2 weeks (CPU)	High: reviewers expect a baseline
3	External test on BraTS-derived 2D slices	2–3 weeks	Very high: answers the main limitation
4	Seeds and 5-fold cross-validation	2–4 weeks (CPU)	Medium
5	DICOM pixel spacing and MedSAM comparison	1–2 months	Second paper

Training runs on this computer take about 13 minutes per epoch and need free memory; close other programs or use a free GPU service (Kaggle, Google Colab) for the baselines and cross-validation.

References

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[12] Capellán-Martín D, et al. Adaptable segmentation pipeline for diverse brain tumors with radiomic-guided subtyping and lesion-wise model ensemble (BraTS 2025). arXiv:2512.14648, 2025. https://arxiv.org/abs/2512.14648

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Preprints [1] and [3] have not been peer reviewed. Their numbers are as reported by the authors.